

A
Project Proposal

On

**“FINAL YEAR PROJECT DESIGN AND DEVELOPMENT OF AN
E-COMMERCE WEB APPLICATION”**

Submitted to

“METROPOLITAN UNIVERSITY, KARACHI”

Karachi

In partial fulfillment for
Bachelors in Computer Science (BSCS)

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Session: **2022 - 2025**

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FINAL YEAR PROJECT PROPOSAL

Title: Design and Development of an E-Commerce Web Application

1. PROJECT TITLE:

Design & Development of a Modern E-Commerce Web Application

2. INTRODUCTION:

This project aims to design and develop a full-featured E-Commerce Web Application that enables users to browse products, place orders, make secure payments, and track deliveries. The system will also include an Admin Dashboard for managing products, orders, customers, and website content.

3. PROBLEM STATEMENT

Traditional retail limits customers by distance and availability, and small businesses lack affordable digital platforms. A secure, user-friendly, and scalable online shopping system is required.

4. OBJECTIVES

Main Objective:

To develop a fully functional and secure web-based e-commerce application.

Specific Objectives:

- Provide a user-friendly product browsing and purchasing interface.
- Implement secure authentication and payment processing.
- Build an admin panel for product, category, and order management.
- Ensure responsive design for different devices.
- Maintain secure data handling and storage.

5. SCOPE OF THE PROJECT

User Side:

- Registration & login
- Product browsing, search, and filters
- Cart & wishlist
- Checkout & payment
- Order tracking
- Ratings & reviews

Admin Side:

- User management
- Product CRUD operations
- Category & inventory management
- Order management

- Sales reports

6. PROPOSED SYSTEM OVERVIEW

Workflow:

1. User browses products.
2. Adds items to cart.
3. Checkout and payment.
4. Order placement.
5. Admin manages stock and orders.

7. TOOLS AND TECHNOLOGIES

Frontend:	HTML, CSS, JavaScript, React.js, Bootstrap/Tailwind
Backend:	Node.js/Express or PHP Laravel
Database:	MySQL or MongoDB
Authentication:	JWT/Sessions
Payment Gateway:	PayPal/Stripe/JazzCash
Version Control:	Git & GitHub

8. METHODOLOGY

Agile / SDLC approach:

- Requirement Analysis
- System Design (ERD, UML)

- UI/UX Design
- Frontend Development
- Backend Development
- Integration & Testing
- Deployment

9. EXPECTED OUTCOMES

- Fully functional e-commerce website
- Secure authentication & payment
- Admin dashboard
- Responsive UI
- Structured database
- Project documentation

10. PROJECT TIMELINE

Requirement Gathering –	1 Week
Design –	1–2 Weeks
Frontend –	2 Weeks
Backend –	3 Weeks
Testing –	1 Week
Final Report –	1 Week
Total Duration:	8–10 Weeks

11. CONCLUSION

The system will serve as a modern digital shopping platform with secure, responsive, and scalable features, suitable as a final year university project.

FINAL YEAR PROPOSAL (FYP-II) M WASEEM AKHTAR & M ALI DANISH BSCS